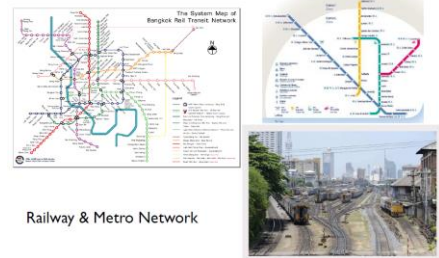


Computer Networks



What is a Network?



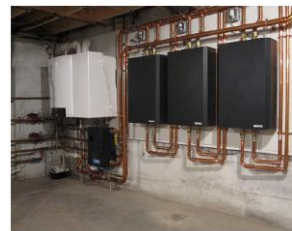
Railway & Metro Network

What is a Network?



Highway Network

What is a Network?



Plumbing Network

What is a Network?

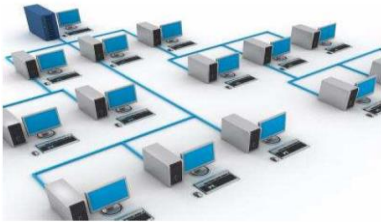


Spider Net(work)

What is a Network?

- A system of lines/channels that are interconnected with each other.
- **What is a Computer Network?**

What is a Computer Network?



What is a Computer Network?

- A set of computers which are connected together.
- This can mean two computers cabled together on the same desk, or thousands of computers across the world.
- The connections can be cables or wireless.



Definitions

- A **network** is a set of devices (often referred to as **nodes**) connected by communication **links**.
- A **node** can be a **host** (such as a computer, a laptop, a smart phone etc.) or a network device (such as a switch, a router, etc.).
- A **link** is a communication pathway that transfer data from one device to another.

Advantages of Computer Networks

- Enables users to **share hardware** like scanners and printers. This reduces costs by reducing the number of hardware items bought.
- Allows users **access to data** stored on others' computers. This keeps everyone up-to-date on the latest data
- Can share access to the **Internet**.

Advantages of Computer Networks

- Can even let users **run programs** that are not installed on their own computers but are installed elsewhere in the network.
 - reduces the effort for networks administrators to keep programs configured correctly and saves a lot of storage space.

Importance of Networks:

- ✓ Easy access and sharing of information
- ✓ Sharing of expensive devices and network resources
- ✓ Modern Technologies (IP telephony, Video Conferencing,etc)

Character Encoding:

- The process of transforming humanly readable character into machine readable code is known as character encoding
- In a particular encoding scheme, characters are turned into a series of 1s and 0s.
- **Why 1s and 0s?**
- The 1s and 0s are used as symbols to represent two discrete states, much like a light switch being ON or OFF.
- These discrete states can be easily represented electrically by discrete voltage of electricity.
- These discrete voltages of electricity coded characters can then be easily transmitted, received and examined by data communications equipment.

13

- The individual 1s and 0s that constitute a given character are known as **bits**.
- The series of **8 bits** representing the entire encoded letter is known as a **byte**.
- These 1s and 0s, or digits, represented by discrete voltages of electricity are in digital format and are known as digital data.
- Now that the humanly readable character is in machine-readable form, it (these bits) can now be transmitted in a process generally known as data communications.

14

Encoding Standards

Characters can be encoded according to a variety of protocols or standards. A few of the currently popular encoding standards are as follows:

- **ASCII American Standard for Information Interchange:**
 - ASCII is one of standard method for encoding humanly readable characters.
 - ASCII - the binary code most widely used with microcomputers.
 - ASCII uses a series of 7 bits to represent 128 ($2^7 = 128$) different characters including uppercase and lower case letters, numerals, punctuation symbols, and specialized control characters.
- **EBCDIC Extended binary coded decimal interchange code:**
 - EBCDIC - used with large computers
 - EBCDIC is an 8 bit code capable of representing 256 different characters, numerals, and control characters. ($2^8 = 256$).
 - EBCDIC is the primary coding method used in IBM mainframe applications.

15

Humanly Readable	ASCII	EBCDIC
A	1000001	11000001
x	1111000	10100111
5	0110101	11110101

Humanly Readable, ASCII, and EBCDIC Coding

16

Unicode and ISO 10646:

- Unicode- uses two bytes for each character rather than one
- ASCII and EBCDIC coding schemes have sufficient capacity to represent letters and characters familiar to people whose alphabets use the letters A, B, C etc.
- ➔ **However, what happens if the computer needs to support communications in Chinese or Arabic?**
- It should be obvious that 128 or 256 possible characters will not suffice when other languages and alphabets are considered.
- In view of this fact, an international effort was undertaken to establish a new coding standard that could support many more alphabets and symbols than ASCII or EBCDIC.
- Although Unicode and ISO 10646 were initiated separately, Unicode version 1.1 and ISO 10646 are identical and were released in 1993.
- Unicode is a 16-bit code supporting up to 65,536 possible characters ($2^{16} = 65536$).

17

The Parity Bit

Parity bit - an extra bit attached to the end of a byte for purposes of checking for accuracy.

- Even parity - sum of bits must come out even
- Odd parity - sum of bits must come out odd

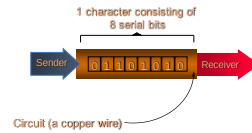
18

Serial Transmission vs Parallel Transmission

	Transmission Description	
	Serial	Parallel
Transmission Description	Bits transmitted in a linear fashion, one bit after another	All bits in a single character transmitted simultaneously.
Comparative Speed	Slower	Faster
Distance limitation	Farther	Shorter
Application	Between two computers, from a computer to an external modem, from computer to relatively slow computer.	Within computer along the computer's busses, from a computer to parallel high speed printers.
Cable Description	All bits travel down a single wire, one bit at a time	Each bit travel down its own wire simultaneously with other bits.

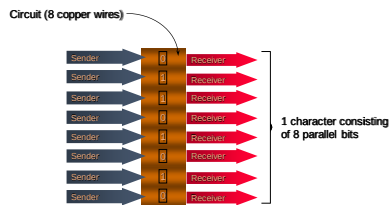
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Serial



20

Parallel

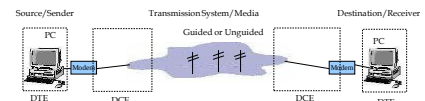


21

Computer Communications and Networks

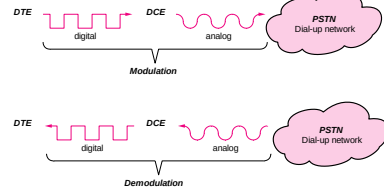
No matter what technology or type of communications is used, three common items are required in order to communicate:

- a source system (Sender)
- a transmission system (Medium)
- a receiver system (Receiver)



A simple communication model

Modulation vs. Demodulation



DCE vs. DTE

- DTE (data terminal equipment) (PC)
- DCE (data communications equipment or data circuit terminating equipment) (modem)

23

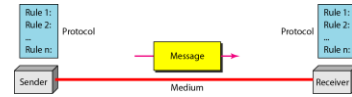
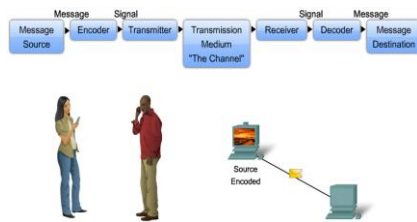
Elements Of Network



Four elements of a network:

- Rules
- Medium
- Messages
- Devices

The elements of communication



28

Classification of Networks

Local Area Network (LAN)
Metropolitan Area Network (MAN)
Wide Area Network (WAN)
Wireless LANs & WANS
Home Networks

Interprocessor distance	Processors located in same	Example
1 m	Square meter	Personal area network
10 m	Room	Local area network
100 m	Building	
1 km	Campus	
10 km	City	Metropolitan area network
100 km	Country	Wide area network
1000 km	Continent	
10,000 km	Planet	
		The Internet

- Classification of interconnected processors by scale.

Local Area Network

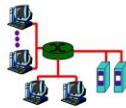
Company/univ **local area network** (LAN) connects end system to edge router

Ethernet:

-shared or dedicated link connects end system and router

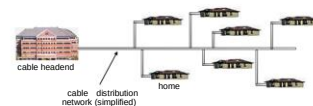
-10 Mbs,
100Mbps, Gigabit Ethernet

Deployment: institutions, home LANs happening now

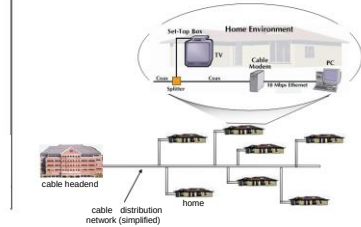


Metropolitan Area Network (MAN)

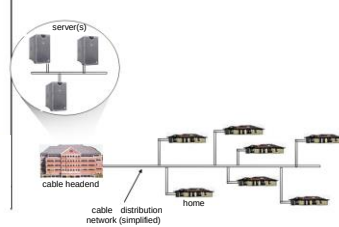
Example : A Cable TV Network



Metropolitan Area Network (MAN)



Metropolitan Area Network (MAN)



• LAN (Local Area Network):

- It is a group of network components that work within small area
 - Short distances
 - Designed to provide local interconnectivity
- Ex.: Token Ring, FDDI, Ethernet, Fast Ethernet, Gigabit Ethernet.....

• MAN (Metropolitan Area Network):

- It is a group of LANs that are interconnected over a small area within city like Jamshoro/Hyderabad/Karachi.
- Provide connectivity over areas such as a city, a campus
- Ex.: Metro Ethernet.

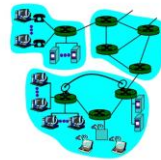
23

Wide Area Network

Spans a large geographic area, e.g., a country or a continent

A WAN consists of several transmission lines and routers

Internet is an example of a WAN



Wireless Networks

shared wireless access network connects end system to router
-via base station aka "access point"

point" wireless

LANs:

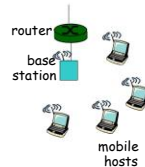
-802.11b (WiFi): 11 Mbps

wider-area wireless

access

4G or 5G

To the wired network

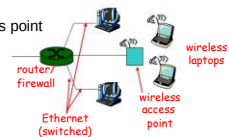


Home Networks

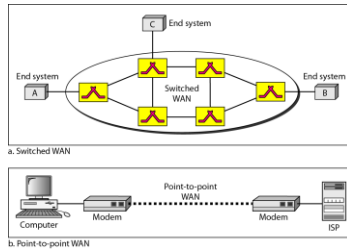
Typical home network components:

router/firewall/NAT
Ethernet

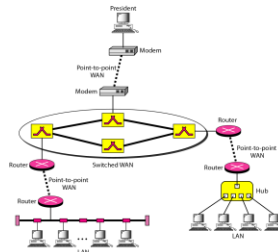
wireless access point



WANS: a switched WAN and a point-to-point WAN



A heterogeneous network made of four WANs and two LANs



BASIC FACTORS

Basic factors that need to be considered in the Computer communication Networks are:

- * The cost of conveying text, images, voice and video messages should be small.
- * The transmission should take place without any iota of doubt/confusion in the mind of the receiver. It means, there should be no interference like noise etc.
- * The message should reach within a reasonable time before it becomes obsolete.
- * For Example: The varying rates of shares/equity in the stock exchange needs to be conveyed immediately so that buyer/seller of shares can decide the action to be taken by them.
- * The message should be safe and secured. It should only reach to the person for whom it is meant.
- * Postal method of conveying a message is not secured at all.
- * Telephonic conversation can be heard if some one taps the telephone line.

39

The objectives of most Computer communication networks are to:

- Reduce the time and efforts required to perform various business tasks.
- Capture business at its source.
- Centralized control over business data.
- Effect rapid dissemination of information .
- Reduce current and future costs of doing business.
- Support business expansion at reasonable incremental cost as the organization grows.
- Support organizational objectives in centralizing or decentralizing computer systems.
- Support improved management control of the organization.

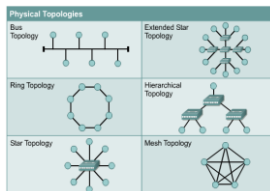
40

Network Topologies

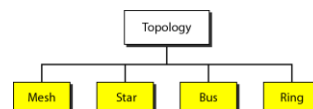
• Topology:

How devices are connected together

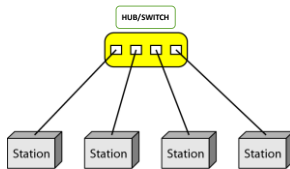
- ✓ **Physical Topology:** It describes how devices are physically cabled
- ✓ **Logical Topology:** It describes how devices communicate across physical topology.



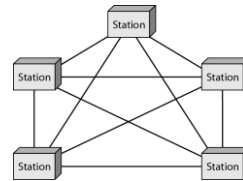
Categories of topology



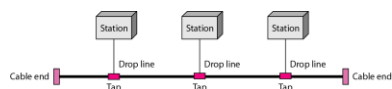
A star topology connecting four stations



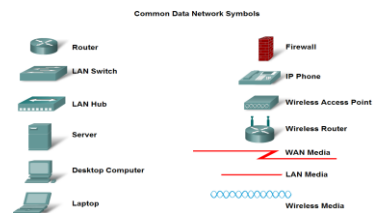
A fully connected mesh topology (five devices)



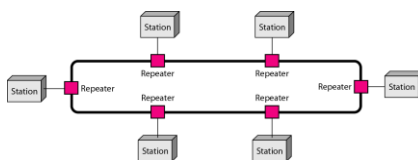
A bus topology connecting three stations



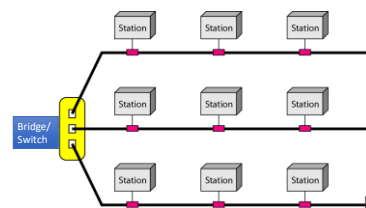
Cisco symbols



A ring topology connecting six stations

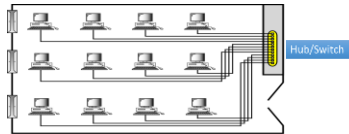


A hybrid topology: a star backbone with three bus networks



An isolated LAN connecting 12 computers to a hub/switch in a closet

Network energy efficiency starts in the LAN closet Network closet component.



Network closet component

energy efficiency can start in the network closet where LAN switches and routers live alongside a host of other power-hungry equipment.

Networks Models

- Computer networks are created by different entities.
- Standards are needed so that these heterogeneous networks can communicate.
- The two most know standards:
 - OSI model: defines a seven layer network.
 - Internet model: defines a five layer network.

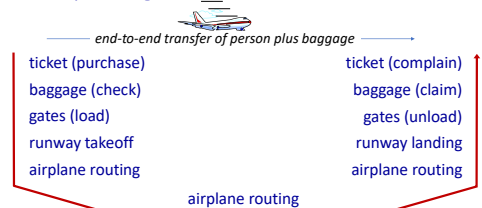
Protocol “layers” and reference models

Networks are complex, with many “pieces”:

- hosts
- routers
- links of various media
- applications
- protocols
- hardware, software

Question: is there any hope of *organizing* structure of network?
▪ and/or our *discussion* of networks?

Example: organization of air travel



How would you *define/discuss* the system of airline travel?
▪ a series of steps, involving many services

Example: organization of air travel

ticket (purchase)	<i>ticketing service</i>	ticket (complain)
baggage (check)	<i>baggage service</i>	baggage (claim)
gates (load)	<i>gate service</i>	gates (unload)
runway takeoff	<i>runway service</i>	runway landing
airplane routing	<i>routing service</i>	airplane routing

layers: each layer implements a service

- via its own internal-layer actions
- relying on services provided by layer below

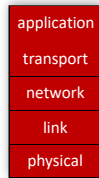
Why layering?

Approach to designing/discussing complex systems:

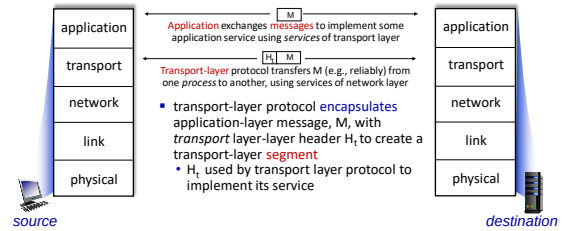
- explicit structure allows identification, relationship of system's pieces
 - layered *reference model* for discussion
- Modularization eases maintenance, updating of system
 - change in layer's service *implementation*: transparent to rest of system
 - e.g., change in gate procedure doesn't affect rest of system

Layered Internet protocol stack

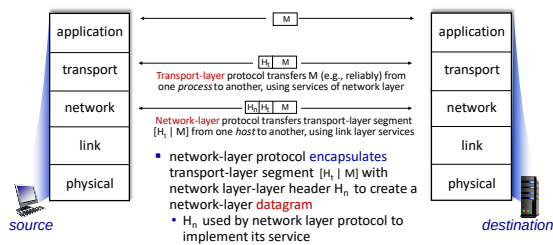
- application:** supporting network applications
 - HTTP, IMAP, SMTP, DNS
- transport:** process-process data transfer
 - TCP, UDP
- network:** routing of datagrams from source to destination
 - IP, routing protocols
- link:** data transfer between neighboring network elements
 - Ethernet, 802.11 (WiFi), PPP
- physical:** bits "on the wire"



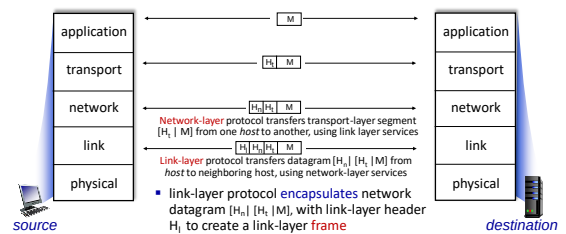
Services, Layering and Encapsulation



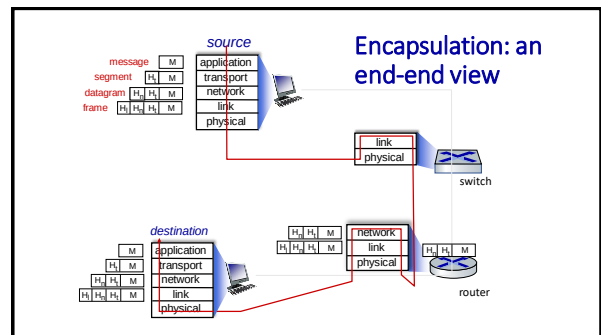
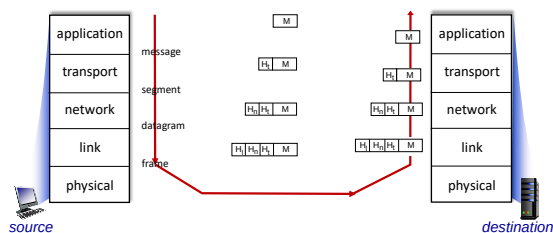
Services, Layering and Encapsulation



Services, Layering and Encapsulation



Services, Layering and Encapsulation



Network Models Summary

OSI Model	Internet Model	Groups of Layers
7. Application layer	5. Application layer	Application Layer
6. Presentation Layer		
5. Session Layer		
4. Transport Layer	4. Transport Layer	Internetwork Layer
3. Network Layer	3. Network Layer	
2. Data link Layer	2. Data link Layer	Hardware Layer
1. Physical Layer	1. Physical Layer	